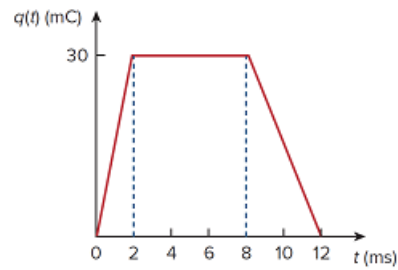


Problem

The charge entering a certain element is shown in Fig. 1.23. Find the current at:

- (a) $t = 1$ ms
- (b) $t = 6$ ms
- (c) $t = 10$ ms

Figure 1.23:



Step-by-step solution

Step 1 of 4

Consider the following waveform for charge entering a certain element:

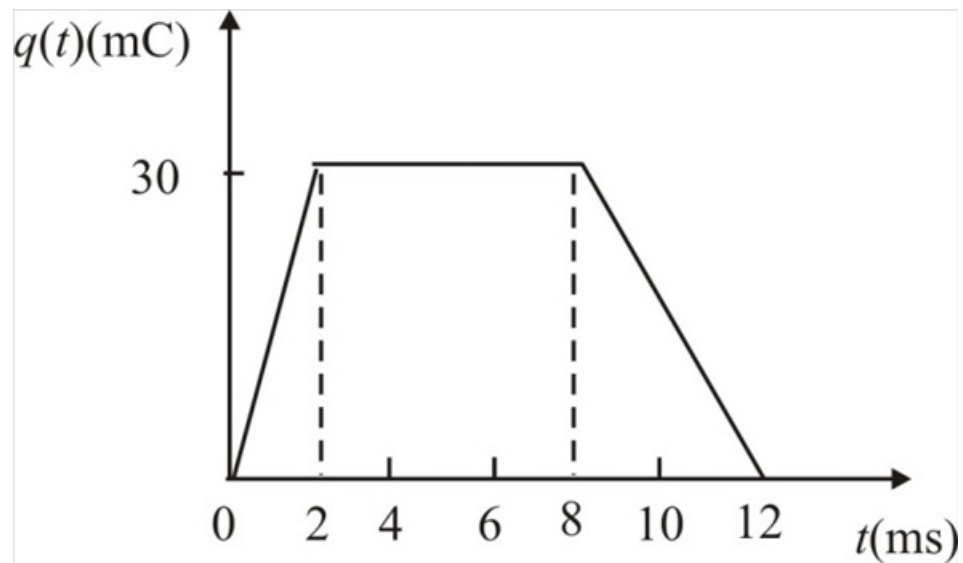


Figure 1

Step 2 of 4

(a)

Calculate the current in the element at $t = 1 \text{ ms}$.

Use the waveform in Figure 1, to write the equation for charge from $t = 0$ to 2 ms .

The response is a straight line between the points $(0, 0)$ to $(2 \text{ ms}, 30 \text{ mC})$. Therefore, write a straight line equation to get an expression for the charge.

$$q - 0 = \left(\frac{30 \text{ mC} - 0}{2 \text{ ms} - 0} \right) (t - 0)$$

$$q = 15t \text{ C}$$

The current is the rate of change of charge measured in amperes.

$$\begin{aligned} i &= \frac{dq}{dt} \\ &= \frac{d}{dt}(15t \text{ C}) \\ &= 15 \text{ A} \end{aligned}$$

Therefore the current from $t = 0$ to 2 ms is 15 A .

Thus, the current at $t = 1 \text{ ms}$ is 15 A.

Step 3 of 4

(b)

Calculate the current in the element at $t = 6 \text{ ms}$.

Use the waveform in Figure 1, to write the equation for charge from $t = 2$ to 8 ms .

The response is a straight line constant at 30 mC . Therefore, the following is the expression for the charge.

$$q = 30 \text{ mC}$$

The current is the rate of change of charge measured in amperes.

$$\begin{aligned} i &= \frac{dq}{dt} \\ &= \frac{d}{dt}(30 \text{ mC}) \\ &= 0 \end{aligned}$$

Therefore the current from $t = 2$ to 8 ms is 0

Thus, the current at $t = 6 \text{ ms}$ is 0.

Step 4 of 4

(c)

Calculate the current in the element at $t = 10 \text{ ms}$.

Use the waveform in Figure 1, to write the equation for charge from $t = 8$ to 12 ms .

The response is a straight line between the points $(8 \text{ ms}, 30 \text{ mC})$ to $(12 \text{ ms}, 0)$.

Therefore, write a straight line equation to get an expression for the charge.

$$q - 30 \text{ mC} = \left(\frac{0 - 30 \text{ mC}}{12 \text{ ms} - 8 \text{ ms}} \right) (t - 8 \text{ ms})$$

$$\begin{aligned} q &= (-7.5 \text{ A})(t - 8 \text{ ms}) + 30 \text{ mC} \\ &= -7.5t + 60 \text{ mC} + 30 \text{ mC} \\ &= -7.5t + 90 \text{ mC} \end{aligned}$$

The current is the rate of change of charge measured in amperes.

$$\begin{aligned} i &= \frac{dq}{dt} \\ &= \frac{d}{dt}(-7.5t + 90 \text{ mC}) \\ &= -7.5 + 0 \\ &= -7.5 \text{ A} \end{aligned}$$

Therefore, the current from $t = 8$ to 12 ms is -7.5 A .

Thus, the current at $t = 10 \text{ ms}$ is $\boxed{-7.5 \text{ A}}$.